Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-II-Q02] | Part A (Q1-Q5)

A cube has vertices $(7, 12, 10), (8, 8, 1), (11, 3, 9)$ among its eight vertices. Find its surface area.

立方体八顶点中包含 $(7, 12, 10), (8, 8, 1), (11, 3, 9)$，求表面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q05] | Part A (Q1-Q5)

Take all points inside or at Euclidean distance at most 1 from a $3 \times 4 \times 5$ solid box. Its volume is $(m + n\pi)/p$ with positive integers and $\gcd(n, p) = 1$. Find $m + n + p$.

取 $3 \times 4 \times 5$ 实心长方体内部及到它欧氏距离不超过 1 的全部点。体积为 $(m + n\pi)/p$，参数为正整数且 n, p 互质，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q04] | Part A (Q1-Q5)

The four face centers of a regular tetrahedron form another tetrahedron. If their volume ratio is m/n reduced, find $m+n$.

正四面体四个面心组成另一个四面体。若小大体积比约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q05] | Part A (Q1-Q5)

A radius-6 circular cylinder is cut by one plane perpendicular to its axis and another making 45° with that plane. The planes' intersection line touches the cylinder at just one point. The wedge between the cuts has volume $n\pi$. Find n.

半径六的圆柱被一垂直轴线的平面及与它成 45° 的另一平面切割，两平面交线与圆柱恰有一个公共点。两切面间楔形体积为 $n\pi$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q05] | Part A (Q1-Q5)

A cone of base radius r and height h rolls on its side without slipping around its fixed vertex. Its first return to the initial position requires seventeen complete spins about its own axis. Express $h/r = m\sqrt{n}$ with squarefree n and find $m+n$.

底半径 r、高 h 的圆锥侧卧无滑动滚动，顶点固定，首次回原位共绕自身轴转十七整周。若 $h/r = m\sqrt{n}$ 且 n 无平方素因子，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-II-Q05] | Part A (Q1-Q5)

A side-15 square is concentric and parallel-sided inside a side-40 square. Join each outer side midpoint to the two nearest inner vertices. Cut out the inner square and four attached triangles, and fold along the inner sides into a square pyramid. Find its volume.

边长十五的正方形与边长四十的外正方形同心且各边平行。将外正方形各边中点连至内正方形最近的两个顶点，剪下内部正方形及四个三角形并折成正四棱锥，求体积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2016-I-Q04] | Part A (Q1-Q5)

A right prism has regular hexagonal bases of edge length 12 and height h. Vertex A and its three neighboring prism vertices form a tetrahedron. The dihedral angle between its base-plane face and its face opposite A is 60°. Find h^2.

直六棱柱的底面为边长 12 的正六边形，高为 h。一个顶点 A 及与它相邻的三个棱柱顶点组成四面体。位于底面内的面与 A 的对面所成二面角为 60°。求 h^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2017-I-Q04] | Part A (Q1-Q5)

A tetrahedron has an isosceles base of side lengths $20, 20, 24$; each edge joining its apex to a base vertex has length25. Its volume is $m\sqrt{n}$ with squarefree positive integer n. Find $m + n$.

四面体的等腰底面三边为 $20, 20, 24$，顶点到底面三个顶点的棱均长25。体积为 $m\sqrt{n}$，n 为无平方因子的正整数。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2000-I-Q08] | Part B (Q6-Q10)

A sealed right circular cone has height 12 and base radius 5. Liquid is 9 deep with the apex down. After turning the apex upward, its depth is $m - n\sqrt[3]{p}$, with positive integers m, n, p and cube-free p. Find $m + n + p$.

密封直圆锥高 12、底面半径 5。尖端向下时液深为 9。倒置为尖端向上后，液深为 $m - n\sqrt[3]{p}$，其中 p 无立方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q06] | Part B (Q6-Q10)

Add the areas of all triangles formed by three distinct vertices of a unit cube. If the result is $m + \sqrt{n} + \sqrt{p}$ for integers m, n, p, find $m + n + p$.

将单位立方体任意三个不同顶点构成的三角形面积全部相加。若结果写为 $m + \sqrt{n} + \sqrt{p}$，其中参数为整数，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2005-II-Q10] | Part B (Q6-Q10)

The face centers of a regular octahedron are the vertices of a cube. If octahedron volume divided by cube volume is reduced m/n, find $m+n$.

正八面体的八个面心构成一个立方体的顶点。八面体与该立方体体积比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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体积与立体几何

[AIME-2012-I-Q08] | Part B (Q6-Q10)

Label a unit cube by $A=(0,0,0), B=(1,0,0), C=(1,1,0), D=(0,1,0)$ and E, F, G, H directly above them. A plane through D and the midpoints of AB, CG divides the cube. If the larger volume is reduced p/q, find $p+q$.

单位立方体的底面顶点为 $A=(0,0,0), B=(1,0,0), C=(1,1,0), D=(0,1,0)$，$E, F, G, H$ 分别在其正上方。过 D 及 AB, CG 中点的平面将其分成两部分，若较大体积为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-II-Q09] | Part B (Q6-Q10)

A cylindrical barrel of radius 4 ft and height 10 ft is full of water. An 8-ft-edge solid cube is lowered centrally into it, with one body diagonal vertical, until the cube rests on the rim. If the displaced water volume is v, find v^2.

半径 4 英尺、高 10 英尺的圆柱桶装满水。边长 8 英尺的实心立方体沿桶轴下放，一条体对角线保持竖直，直至卡在桶口。若排水体积为 v，求 v^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2021-I-Q06] | Part B (Q6-Q10)

From one cube vertex A, the adjacent vertices are B, C, D, and the opposite vertex is G. A point P has distances $PB = 60\sqrt{10}, PC = 60\sqrt{5}, PD = 120\sqrt{2}, PG = 36\sqrt{7}$. Determine PA.

立方体中，与顶点 A 相邻的三点为 B, C, D，对顶点为 G。一点 P 满足 $PB = 60\sqrt{10}, PC = 60\sqrt{5}, PD = 120\sqrt{2}, PG = 36\sqrt{7}$，求 PA。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q12] | Part C (Q11-Q15)

Start with a volume-1 regular tetrahedron $\mathcal{P}_0$. To form each next solid, attach an outward regular tetrahedron on the midpoint triangle of every triangular face, replacing that midpoint patch. If $\mathrm{vol}(\mathcal{P}_3) = m/n$ reduced, find $m+n$.

从体积为 1 的正四面体 $\mathcal{P}_0$ 出发。每次在每个三角面中点三角形上向外附加正四面体，替换原中部面片，得到下一个立体。若 $\mathcal{P}_3$ 体积约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q15] | Part C (Q11-Q15)

A cube of side 8 has opposite vertices A, E. On the three edges from E, take I, J, K at distance 2 from E. Drill the triangular tunnel bounded by the planes through IJ, JK, KI parallel to AE. The remaining solid has total surface area, including tunnel walls, $m + n\sqrt{p}$ with squarefree p. Find $m + n + p$.

棱长为 8 的立方体有对顶点 A, E。在 E 出发的三条棱上取 I, J, K，均距 E 为 2。以过 IJ, JK, KI 且平行体对角线 AE 的三个平面为壁钻出三角隧道。剩余立体包括隧道壁的总表面积为 $m + n\sqrt{p}$，p 无平方因子，求 $m + n + p$。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-I-Q15] | Part C (Q11-Q15)

A six-faced polyhedron has faces $ABCD, ABFG, CDE, ADEG, BCEF, EFG$. The first is a square of side 12; $ABFG$ is an isosceles trapezoid with $AB \parallel GF$, $AG = BF = 8$, $GF = 6$; and $CE = DE = 14$. The distance from E to square face $ABCD$ is 12. Given $EG^2 = p - q\sqrt{r}$ with positive integers and squarefree r, find $p + q + r$.

六面多面体的面为 $ABCD, ABFG, CDE, ADEG, BCEF, EFG$。其中 $ABCD$ 为边长 12 的正方形；$ABFG$ 为等腰梯形，$AB \parallel GF, AG = BF = 8, GF = 6$；$CE = DE = 14$。$E$ 到正方形面 $ABCD$ 的距离为 12。若 $EG^2 = p - q\sqrt{r}$，各参数为正整数且 r 无平方因子，求其和。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q11] | Part C (Q11-Q15)

A height-4, radius-3 cone is painted over its lateral surface and base, then cut parallel to the base into a small cone C and a frustum F. The ratio of their originally painted areas equals their volume ratio, both $k = m/n$ reduced. Find $m + n$.

高四、底半径三的圆锥原侧面与底面全部涂漆，再平行底面切成小圆锥 C 和圆台 F。两者原涂漆面积比等于体积比，均为最简 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q14] | Part C (Q11-Q15)

A taut rope of length 20 runs from a ground-level point on a radius-8 vertical cylinder, around its wall, then to an endpoint at height 4 and horizontal distance 4 from the cylinder surface. Its wall-contact length is $(a-\sqrt{b})/c$, where a, b, c are positive integers and c is prime. Find $a+b+c$.

长二十的绷紧绳从半径八的竖直圆柱底部地面点出发，绕柱壁后伸向高四、水平距柱面四的端点。贴柱壁段长为 $(a-\sqrt{b})/c$，参数为正整数且 c 为质数，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-II-Q11] | Part C (Q11-Q15)

A cone has base radius 600 and height $200\sqrt{7}$. Two surface points lie on opposite generators and are at distances 125 and $375\sqrt{2}$ from the apex. Find their shortest surface-path distance.

圆锥底半径六百、高 $200\sqrt{7}$。两侧面点位于正相对母线上，距顶点分别为 $125, 375\sqrt{2}$，求沿侧面的最短路长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2007-I-Q13] | Part C (Q11-Q15)

A square pyramid has all eight edges length four. A plane passes through the midpoints of one lateral edge AE and base edges BC, CD. If its cross-sectional area is $\sqrt{p}$, find p.

正方锥的八条棱都长四。过侧棱 AE、底边 BC, CD 的中点作平面，截面面积为 $\sqrt{p}$，求 p。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q15] | Part C (Q11-Q15)

From each corner of a side-100 square, mark points $\sqrt{17}$ along its two edges and cut segments from those points to a common point on the inward diagonal, with angle 60° between the cuts. Fold along lines joining neighboring inner cut vertices and tape each pair of cut edges together to form a tray. Its height is $\sqrt[n]{m}$ with $m < 1000$ free of nth-power prime factors. Find $m + n$.

边长一百的正方纸每角沿两边各取距角 $\sqrt{17}$ 的点，向内对角线上同一点切两刀，两切线夹角六十度。沿相邻内切点连线折起，并粘合各角切口，形成托盘。高度为 $\sqrt[n]{m}$，$m < 1000$ 且无素数的 n 次方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2010-I-Q11] | Part C (Q11-Q15)

Revolve the region $|8-x|+y \le 10, 3y-x \ge 15$ about the line $3y-x=15$. Its volume is $m\pi/(n\sqrt{p})$, with m, n coprime and p squarefree. Find $m+n+p$.

将满足 $|8-x|+y \le 10, 3y-x \ge 15$ 的平面区域绕直线 $3y-x=15$ 旋转。体积为 $m\pi/(n\sqrt{p})$，m, n 互质，p 无平方因子。求 $m+n+p$。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-I-Q15] | Part C (Q11-Q15)

A solid right circular cylinder has radius 6 and height 8. Two points A, B on the rim of one base cut off a minor arc of 120°. A plane through A, B and the cylinder center produces a flat cross-section of area $a\pi + b\sqrt{c}$, with integral a, b, c and squarefree c. Find $a + b + c$.

实心直圆柱半径为 6、高为 8。同一底面边缘上两点 A, B 所夹小弧为 120°。过 A, B 和圆柱中心的平面截得平面图形，其面积为 $a\pi + b\sqrt{c}$，其中 a, b, c 为整数，c 无平方因子。求 $a + b + c$。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-I-Q13] | Part C (Q11-Q15)

Two noncongruent parallelepipeds have all six faces rhombi with diagonal lengths $\sqrt{21}, \sqrt{31}$. If their larger-to-smaller volume ratio is the reduced fraction m/n, find $m+n$.

两个不全等平行六面体的每个面都是对角线长为 $\sqrt{21}, \sqrt{31}$ 的菱形。较大体积与较小体积之比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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体积与立体几何

[AIME-2023-II-Q14] | Part C (Q11-Q15)

A cube has parallel edges AB, CD and face diagonals AC, BD. The rectangle $ABDC$ lies in a vertical plane. The heights of A, B, C, D above a horizontal floor are $0, 2, 8, 10$ meters. Water fills the cube up to height 7 meters. If its volume is m/n cubic meters in lowest terms, find $m + n$.

立方体的 AB, CD 为平行棱，AC, BD 为面对角线，矩形 $ABDC$ 所在平面竖直。A, B, C, D 离水平地面的高度分别为 $0, 2, 8, 10$ 米。水面高度为 7 米。若水量为最简分数 m/n 立方米，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2024-I-Q14] | Part C (Q11-Q15)

Tetrahedron $ABCD$ has opposite-edge pairs $AB = CD = \sqrt{41}$, $AC = BD = \sqrt{80}$, $AD = BC = \sqrt{89}$. Its interior point equidistant from all faces has distance $m\sqrt{n}/p$, with coprime m, p and squarefree n. Find $m + n + p$.

四面体 $ABCD$ 的对棱分别满足 $AB = CD = \sqrt{41}, AC = BD = \sqrt{80}, AD = BC = \sqrt{89}$。内部到四面等距的点，其共同距离为 $m\sqrt{n}/p$，其中 m, p 互质且 n 无平方素因子。求 $m + n + p$。

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